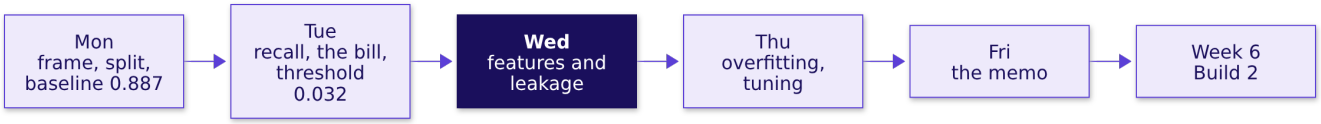
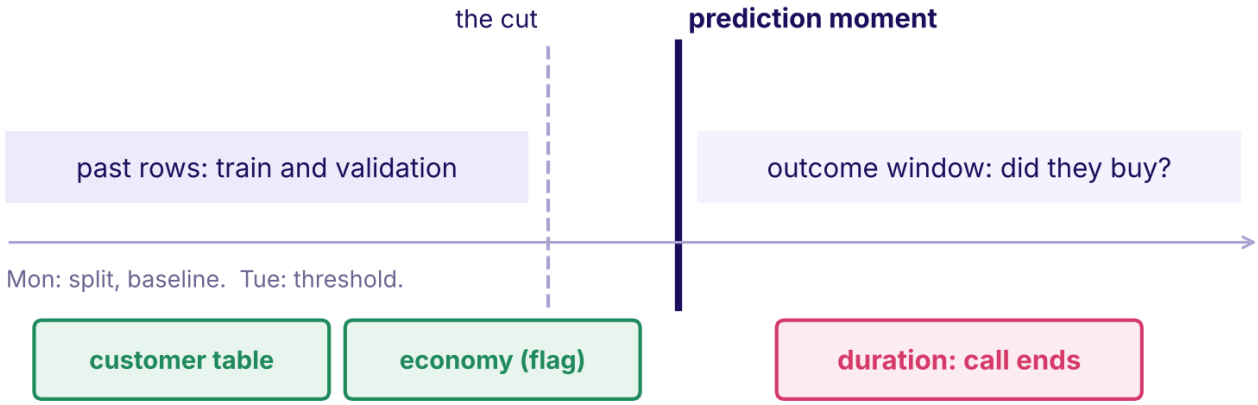
Week 5 Wednesday study notes: what the model is allowed to know

Where does Wednesday sit?



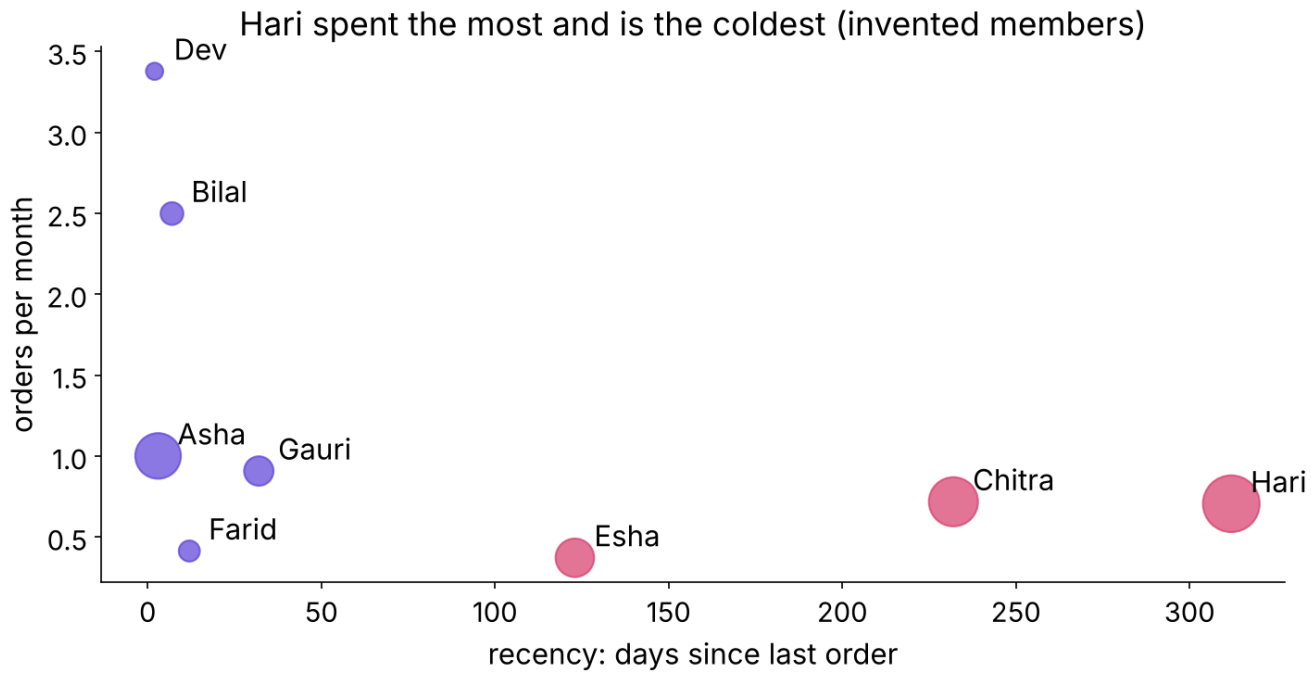
Monday decided what is predicted and when; Tuesday decided how to score it in rupees. Wednesday decides what the model may read. Thursday tunes on Wednesday's audited feature set, so every choice made today travels.



The one sentence to keep: **on the morning the model is used, do we know this value for this customer?**

The five topics, in the curriculum's words	The chapter
Feature engineering from the customer table: ratios, recency, counts, cohort age, basket signals	1 Turning raw columns into signals
Encoding categoricals; scaling numerics; which models care	2 Teaching a model to read categories
Leakage and the prediction-time question	3 Answers from the future
The column audit	4 Interrogating every column
Feature importance and its limits	5 What the importance chart can't tell you

Chapter 1: how does a raw column become a signal?



The idea in three lines. A raw total grows with time, so it ranks the oldest customers first. A signal is measured from the prediction moment and puts every customer on one scale. Each signal carries a business reason a marketing lead would repeat.

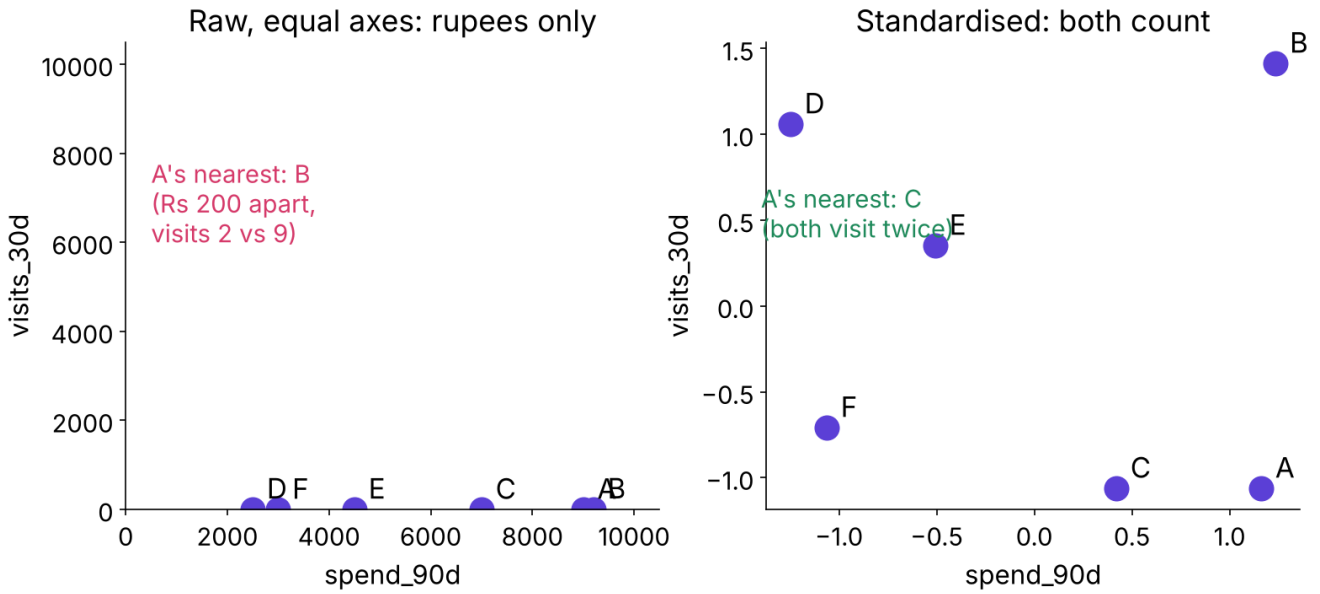
The worked example (invented). On the nudge morning, 1 November 2026, Hari has the largest total spend (Rs 26,000) and last ordered 312 days ago. Asha's average basket is Rs 16,800 over 24 orders, Rs 700; Dev's recency is 2 days and he orders 3.38 times a month as a member.

Signal	Formula	The reason
Recency	nudge morning - last order, in days	warm buys again
Count	orders in the last 90 days	habit
Cohort age	nudge morning - first order	new members are still deciding
Ratio	orders / months as a member; spend / orders	old and new on one scale
Basket signal	items / orders	weekly shop against top-up

The trap. `pd.Timestamp.today()` in place of the nudge morning: recency then counts days that had not happened when the nudge went out. In the bank's file, ringing a client more often this campaign lowers the yes-rate, from 13 percent at one call to 7 percent at four or more, which is why each feature gets a check as well as a reason.

The interview line. "Every feature I build is anchored to the prediction moment and carries a reason the business would say aloud."

Chapter 2: how does a model read a category or a rupee?



The idea in three lines. A label code (Pune 1, Surat 2, Indore 3) hands a linear model an order and equal gaps nobody chose; one-hot gives each city its own weight. Rupees drown visits in any distance until both are standardised. Trees split on thresholds, so they shrug at both.

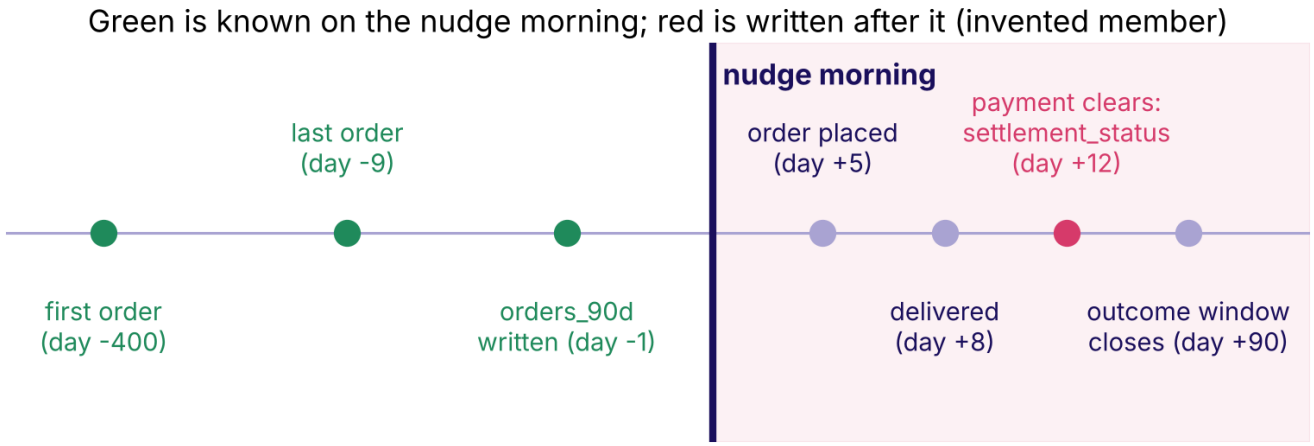
The worked example (invented). Member A against member B: spend differs by Rs 200 and visits by 7, so the raw distance is the square root of 200 squared plus 7 squared, about 200.1, and B is A's nearest neighbour. After `z = (value - mean) / standard deviation`, A's nearest neighbour is C, who also visits twice a month.

Model family	Category	Number
Logistic, linear	one-hot	standardise
Nearest neighbours, clustering	one-hot	standardise, decisive
Tree, forest, boosting	one-hot or an ordinal code	no need

The trap. Fitting the scaler on all rows before the split. The bank's means move only in the third decimal, which is why the habit looks harmless; the rule holds anyway, because on a file where the future differs it carries the future in.

The interview line. "Encoding and scaling are decisions about the model family, made inside the pipeline on training rows."

Chapter 3: what is leakage, and which question catches it?



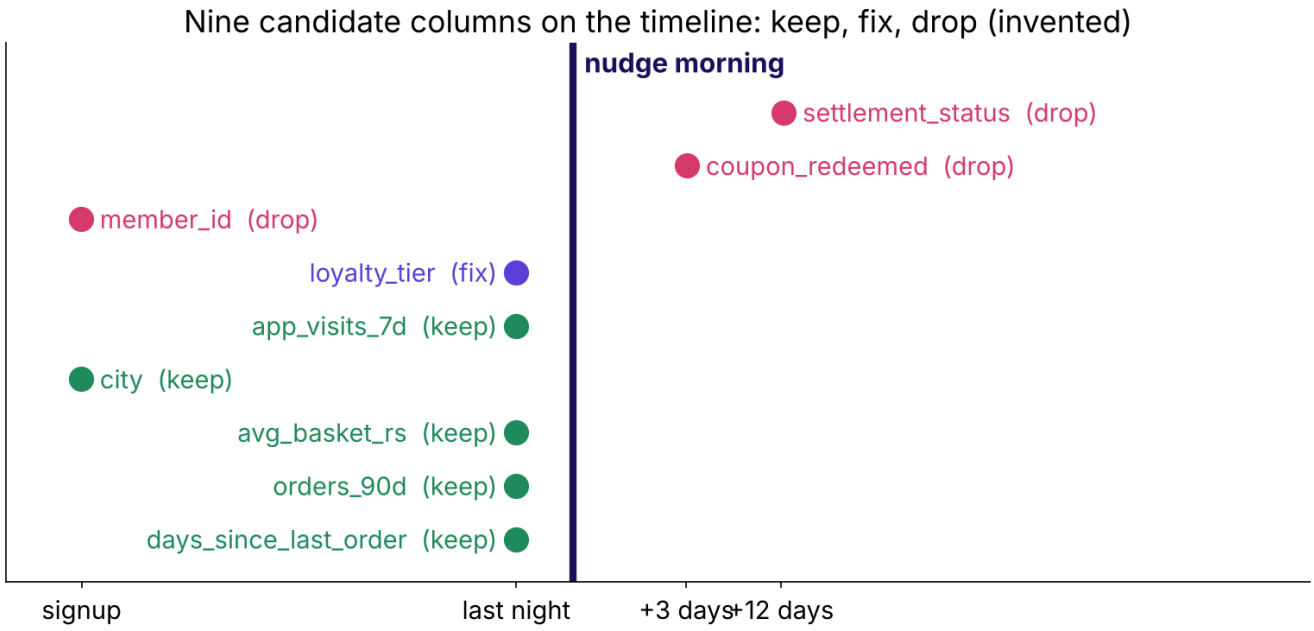
The idea in three lines. Leakage is any input the model sees in training that would not be known at the moment it is used. The prediction-time question catches it when asked of every column. A column written after the moment is dropped, however well it scores.

The worked example (invented). Twelve Kalpa members. "Bought if `settlement_status` is settled" gets 12 of 12 on the training table, built after the outcome. On the nudge morning no payment from the coming 90 days exists, the column reads "none" for everyone, and the rule falls to 6 of 12, the score of predicting nobody buys. The best honest rule, two or more orders in 90 days, gets 10 of 12.

The trap. Reading "too good" as the verdict. In the bank's file `poutcome = success` converts at 65 percent and stays, because the earlier campaign closed months before the call.

The interview line. "Leakage is a question about time: when is the column written, against the moment the model runs?"

Chapter 4: how do you audit every column?



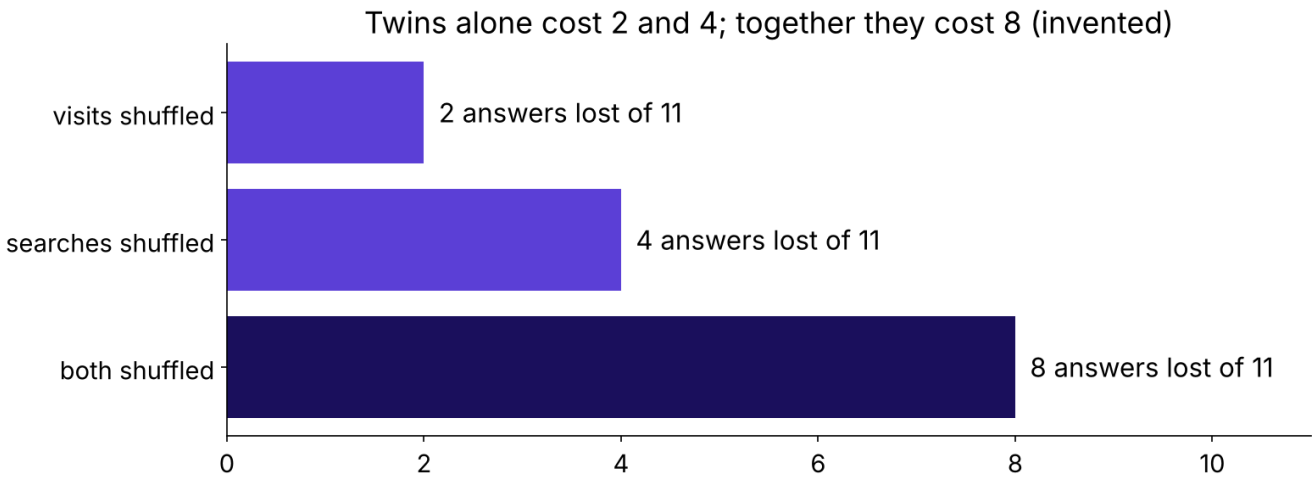
The idea in three lines. Five asks per column: who writes it, when, is it overwritten, its business reason, the verdict. Overwritten fields are read as a snapshot from the prediction morning. Identifiers and columns with no reason are dropped.

The worked example (invented). Of nine candidates, `coupon_redeemed` (day +3) and `settlement_status` (day +12) are written after the nudge, `member_id` has no reason, and `loyalty_tier` is kept from a snapshot, because member M07 bought, was upgraded to Gold, and today's table no longer shows the Silver of the nudge morning.

The trap. "If it is in the warehouse, it was known." The warehouse holds today's value.

The interview line. "I audit the boring columns first, with the owner, one line per column."

Chapter 5: what can an importance chart prove?



The idea in three lines. Permutation importance shuffles a column and measures how far the score falls: how much this fitted model leans on it. Close partners share credit, so they are shuffled and read as a group. Importance names a lever only after a fair test changes the column.

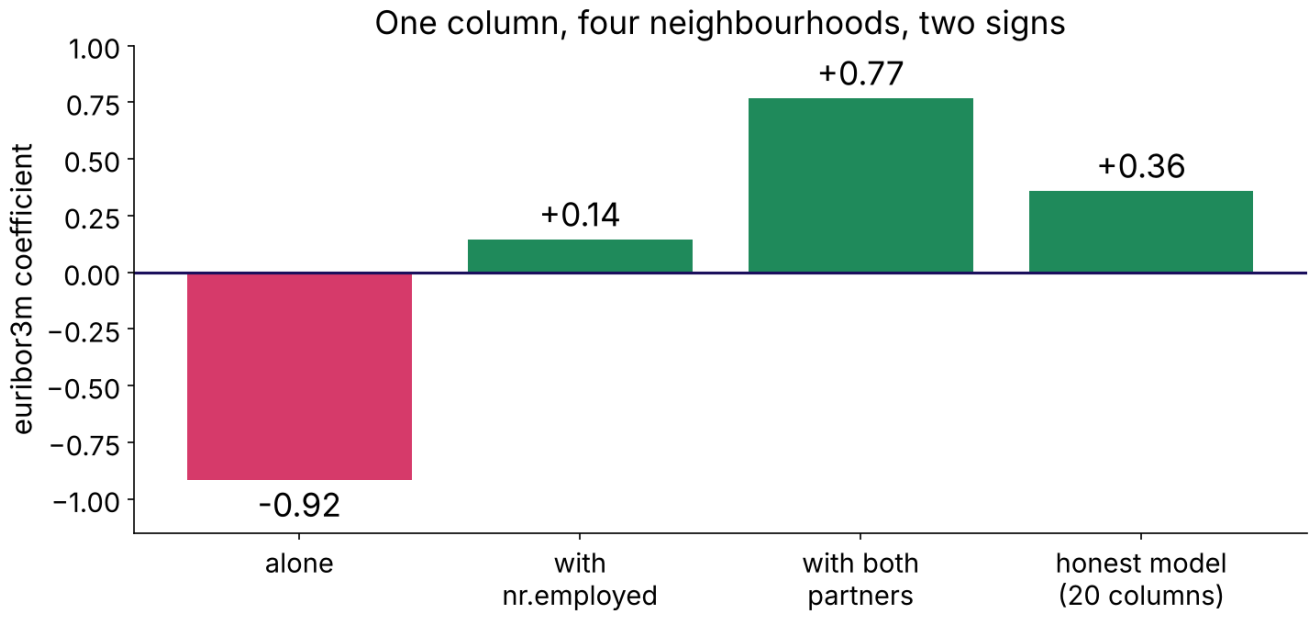
The worked example (invented). A rule (visits plus searches of 8 or more) gets 11 of 12. Shuffling visits costs 2 answers, searches 4, both together 8, more than the two single costs added, because the twins correlate at 0.94.

The trap. Reading the top bar as the driver. Members search because they already intend to buy; pushing searches up would not create that intent.

The interview line. "Importance ranks reliance. Cause needs an experiment."

What did the bank's 41,188 calls show?

Step	Number	What it means
Customer table	AUC 0.761	Monday's model
+ economy	AUC 0.795	most of the gain is the calendar
+ call log (`duration`)	AUC 0.932, accuracy only 0.899 to 0.907	the jump that should start the doubt
`duration` by length	0.5 percent yes in the shortest fifth, 33 in the longest	the call is the sale
Year and month alone	AUC 0.785	the calendar leaks through a random split
Honest set, inside one month	AUC about 0.57	the ranking a call list really gets
The bill at Rs 40 a call, Rs 1,200 a missed buyer	leak Rs 1.19 lakh, honest Rs 2.82 lakh	the promise no live list could keep
`euribor3m`	-0.92 alone, +0.77 beside two partners	read twins as a group
`emp.var.rate` shuffled alone	0.34 AUC lost against a lift of 0.30	the shuffle built clients who never existed



The dataset's authors write that a call's length strongly shapes the outcome, is unknown before dialling, and belongs in benchmarks only (UCI Bank Marketing page, https://archive.ics.uci.edu/dataset/222/bank+marketing, verified 28 October 2026).

Which real teams met the same problems?

Team	What happened	Source, verified 28 October 2026
Zech and others, chest X-rays, PLOS Medicine 2018	A pneumonia model scored internal AUC 0.931 and 0.815 at a hospital it had never seen; the network identified the hospital system of 99.95 percent of NIH images, partly from a metal token, and hospital prevalence alone ranked at AUC 0.861	https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.1002683 (verified 28 October 2026)
Kaggle IJCNN 2011 social network challenge, Narayanan, Shi and Rubinstein	The winners re-identified much of the anonymised test set from their own Flickr crawl and reached AUC 0.981, largely by looking up answers	https://arxiv.org/abs/1102.4374 (verified 28 October 2026)
Airbnb search, Haldar and others	Their first neural network stalled because features were not normalised; distribution plots then exposed a monthly price logged as daily for stays over 28 days	https://arxiv.org/abs/1810.09591 (verified 28 October 2026)
Uber Michelangelo, 2017	UberEATS delivery time is predicted from request, seven-day and last-hour features, kept in a store where "the same expressions are applied at training time and at prediction time"	https://www.uber.com/blog/michelangelo-machine-learning-platform/ (verified 28 October 2026)

The Zech case is the bank's calendar in another field: the hospital, like the month, is known when the prediction is made and still stands in for a base rate that differs at the next site.

How does each seat look at Kalpa's 0.99?

Seat	The question	Today's answer
Business owner	What will the nudges cost, and what will they return?	only the honest model's bill can be collected
Product manager	What does the member see, and when?	a nudge on the morning, before any order
Solution architect (spotlight)	Is the column there on the morning, in the feed the service reads?	`settlement_status` arrives days later: drop
Senior ML engineer	Is the jump believable?	one column worth 0.99 is a leak until its write time is known
Implementation engineer (spotlight)	Where are the encoder and the scaler fitted?	inside the pipeline, on training rows only
QA and tester	Which test fails if a late column slips in?	an audit check that every column's write day is before the moment

What do you read or watch next?

Link	Why
scikit-learn, Common pitfalls (verified 28 October 2026): https://scikit-learn.org/stable/common_pitfalls.html	leakage in preprocessing and the pipeline fix
scikit-learn, Permutation importance with multicollinear features (verified 28 October 2026): https://scikit-learn.org/stable/auto_examples/inspection/plot_permutation_importance_multicollinear.html	the twin problem on a real example
Google Machine Learning Crash Course, One-hot encoding (verified 28 October 2026): https://developers.google.com/machine-learning/crash-course/categorical-data/one-hot-encoding	encoding from zero
CampusX, What is Feature Engineering (video, verified 28 October 2026): https://www.youtube.com/watch?v=sluoVhT0ehg	chapter 1 as a short lecture
Normalized Nerd, Standardization vs Normalization (video, verified 28 October 2026): https://www.youtube.com/watch?v=sxEqtjLC0aM	chapter 2's scaling picture
A Data Odyssey, Permutation Feature Importance from Scratch (video, verified 28 October 2026): https://www.youtube.com/watch?v=U-iahfXlSoY	chapter 5 coded by hand

The IITGN faculty session W5-3 (tentative) picks up from here with multicollinearity, variance inflation and target encoding.

Can you answer these without notes?

1. State the prediction-time question word for word.
2. Why does accuracy barely move when AUC jumps on an imbalanced file?
3. Name two ways a column known in time can still mislead a score.
4. Which model families shrug at unscaled features, and why?
5. What does a single-column importance larger than the model's lift tell you?